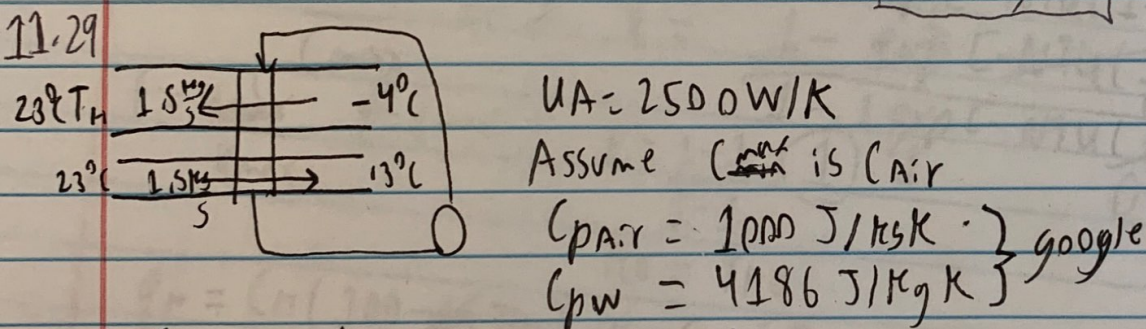


Matt Egeley
HW 9

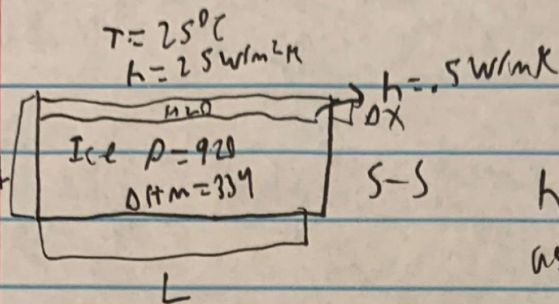
11.41 $U_{\text{clean}} = 300 \text{ W/m}^2\text{K}$ $U_{\text{dirty}} = (.75)(U_{\text{clean}}) = 225 \text{ W/m}^2\text{K}$
 $R''_{fi} = q_i t = (2.5 \times 10^{-11}) +$ $R''_{fo} = q_o t = (1.0 \times 10^{-11}) +$
 $\frac{1}{UA} = \frac{R''_{fi}}{A} + \frac{R''_{fo}}{A}$ ~~$\frac{1}{U} = R''_{fi} + R''_{fo}$~~ $\frac{1}{U} = R''_{fi} + R''_{fo} = (2.5 \times 10^{-11}) + (1.0 \times 10^{-11}) +$
 $U = 3.5 \times 10^{-11} +$
 $\frac{1}{U} = 3.5 \times 10^{-11} + = \frac{1}{U_{\text{dirty}}} - \frac{1}{U_{\text{clean}}}$
 $3.5 \times 10^{-11} + = \frac{1}{225} - \frac{1}{300} = .0011$
 $+ = .0011 / 3.5 \times 10^{-11}$
 $= 3.17 \times 10^7 \text{ s} \approx \boxed{367 \text{ days}}$ 1 year to make it easy



$C_{\text{max}} = 1.5 \cdot 1000 = 1500 \text{ W}$
 $q_{\text{air}, B} = 1500 (+23 - 13) = +15,000 \text{ W}$
 $q_{\text{w}, O} = (W (T_{\text{w}, O} - T_{\text{w}, i}))$ $15,000 = (W (T_{\text{w}, O} - T_{\text{w}, i}))$
 $q_{\text{air}, A} = 1500 (+23 - (-4)) = +40,500$
 $q_{\text{w}, A} = (W (T_{\text{w}, O} - T_{\text{w}, i}))$ $T_{\text{w}, O} = T_{\text{w}, i}$
 $40,500 = (W (T_{\text{w}, O} - T_{\text{w}, i}))$
 $(W (T_{\text{w}, O} - T_{\text{w}, i})) - 15,000 = (W (T_{\text{w}, O} - T_{\text{w}, i})) - 40,500$

?

Exam A)
P1



$$E_{in} = E_{out}$$

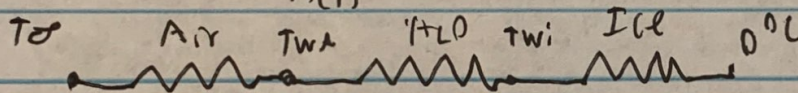
heat released from melting ice
as function of time

$$q = m \Delta H_m \quad m = \rho L^2 \frac{dx}{dt}$$

$$= \rho L^2 \Delta H_m \frac{dx}{dt}$$

heat transferred through ice & water

$$q = \frac{k(T_{wi} - T_{wa})L^2}{X(t)} = h(T_{wa} - 25)L^2$$



$$q_{ice\ melt} = T_{\infty} - T_{wi}$$

$$\frac{X}{k_w} + \frac{1}{hA}$$

$$\rho L^2 \Delta H_m \frac{dx}{dt} = \frac{T_{\infty} - T_{wi}}{\frac{X}{k_w} + \frac{1}{hA}}$$

$$q \Delta H_m \frac{dx}{dt} = \frac{T_{\infty} - T_{wi}}{\frac{X}{k_w} + \frac{1}{hA}}$$

Exam $\rightarrow 100^\circ\text{C}$ 65°C $q_H = C_H (100 - 65)$

PZ $\rightarrow 20^\circ\text{C}$ $?^\circ\text{C}$ $q_L = C_L (20 - T_{CO})$

Same fluid $C_H = C_L$

$$(100 - 65) = -(20 - T_{CO})$$

$$35 = -20 + T_{CO} \quad T_{CO} = \boxed{55^\circ\text{C}}$$

B) $\rightarrow 100$ 65
 $?$ $20 \leftarrow$

From eq for counter flow

$$\epsilon = \frac{1 - \exp[-NTU(1-r)]}{1 - r \exp[-NTU(1-r)]}$$

$C_r = \frac{C_{\min}}{C_{\max}} = 1$ $\epsilon = \frac{1 - \exp[-NTU(1-1)]}{1 - 1 \exp[-NTU(1-1)]}$

$\epsilon = \frac{q}{q_{\max}}$ $q = q_{\max}$

$\epsilon = \frac{1-1}{1-1} = \frac{0}{0}$

$T_{CO} = T_{HO}$

$q_H = C_H (100 - T_{HO})$ $q_L = C_L (T_{CO} - 20)$

$T_{CO} - 20 = 100 - T_{HO}$

$2T_{HO} = 120 \quad T_{HO} = \boxed{60^\circ\text{C}} = T_{CO}$

Comment Summary